Introduction

The choice of distance or dissimilarity is the foundation of most multivariate methods: clustering algorithms (k-means, hierarchical, PAM, DBSCAN) consume a distance, multidimensional scaling and PCoA produce coordinates that approximate one, and k-nearest neighbours computes them at every prediction. Different metrics capture different notions of proximity, and the right choice depends on the data type — continuous vs. categorical, abundances vs. presence/absence, magnitudes vs. directions — and on the inferential question. Mismatched metrics produce results that look correct but answer the wrong question.

Prerequisites

A working understanding of vectors, the difference between distance and dissimilarity (metric vs. semi-metric), and the most common multivariate methods that consume distance matrices.

Theory

Standard continuous distances between p-dimensional vectors:

- **Euclidean**: $\sqrt{\sum_i (x_i - y_i)^2}$ — straight-line distance, the default for most uses.
- **Manhattan (L_1)**: $\sum_i |x_i - y_i|$ — robust to outliers, appropriate for grid-like geometry.
- **Minkowski (L_p)**: $(\sum_i |x_i - y_i|^p)^{1/p}$ — generalises Euclidean ($p = 2$) and Manhattan ($p = 1$).
- **Canberra**: $\sum_i |x_i - y_i|/(|x_i| + |y_i|)$ — sensitive to small values; useful for compositional data.
- **Cosine**: $1 - (x \cdot y)/(\|x\|\|y\|)$ — measures direction, not magnitude; standard in text and gene-expression contexts.
- **Mahalanobis**: scale- and correlation-aware, weights the deviation by the inverse covariance matrix.

For categorical or mixed data: Gower distance handles mixed types via per-variable scaling; Jaccard and Bray-Curtis suit presence-absence and abundance data respectively.

Assumptions

Metrics satisfy the triangle inequality and symmetry; dissimilarities may relax these. Some downstream methods (classical MDS, k-means) implicitly assume metric distances; others (NMDS, hierarchical clustering with non-Euclidean distances) handle dissimilarities gracefully.

R Implementation

```r
library(cluster)

x <- rbind(c(1, 2, 3), c(2, 4, 5), c(5, 0, 1))

dist(x, method = "euclidean")
dist(x, method = "manhattan")
dist(x, method = "minkowski", p = 3)
dist(x, method = "canberra")

# Gower for mixed data
d <- data.frame(num = 1:3, cat = c("A", "B", "A"))
daisy(d, metric = "gower")
```

Output & Results

`dist()` returns a `dist` object containing the lower-triangular pairwise distance matrix. Different methods produce numerically different matrices that emphasise different features of the geometry. `cluster::daisy()` extends this to mixed numeric-categorical data via Gower distance.

Interpretation

A reporting sentence: "Euclidean distances of 3.0 between rows 1 and 2 and 5.7 between rows 1 and 3 reflect continuous-valued separation; the Gower distance on mixed data scaled each variable to $[0, 1]$ before combining and produced a unitless dissimilarity in the same range." Always state the metric used; clustering or ordination output is uninterpretable without it.

Practical Tips

- Standardise variables (`scale()`) before Euclidean distance whenever scales differ across columns; otherwise large-variance variables dominate the distance and bias every downstream method.
- For mixed numeric and categorical data, Gower is the standard; it scales each variable to $[0, 1]$ via type-specific rules and combines them as a weighted average.
- For binary presence/absence, prefer Jaccard or Sørensen-Dice (which ignore joint absences) over Hamming (which counts all matches).
- For sparse high-dimensional data (text, gene expression), cosine distance is often more meaningful than Euclidean because magnitudes vary widely while direction encodes the signal.
- Mahalanobis is essential when predictor correlations matter; it underlies the discriminant function in LDA and Hotelling's T^2.
- Always match the metric to the data type and downstream method; Euclidean on mixed scales or on presence/absence data is a frequent mistake that invalidates clustering results.

R Packages Used

Base R `dist()` for the standard continuous metrics; `cluster::daisy()` for Gower distance; `vegan::vegdist()` for ecological distances (Bray-Curtis, Jaccard, Hellinger, Canberra); `philentropy::distance()` for an extensive catalogue of distance and divergence functions; `proxy::dist()` and `parallelDist::parDist()` for multi-core distance computation on large matrices.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- PCA, factor analysis, CCA, LDA
- Clustering: k-means, hierarchical, model-based
- UMAP and t-SNE